

Firm-level price wedges: how good is the mapping, and can it be improved?

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What is being measured

A firm's price wedge is not estimated directly. Wedges are estimated for decile portfolios, and a mapping carries those onto individual firms through their characteristics. The question is how much that mapping can be trusted, and whether it can be improved.

Four criteria, because they answer different questions.

Leaving one characteristic out. Fit on the portfolios of all but one characteristic, predict that one's ten deciles, rotate. Can the mapping price an anomaly it has not seen?

Leaving one family out. The same, holding out a whole Freyberger–Neuhierl–Weber category. Harder and more honest: value characteristics predict each other, so holding out book-to-market while assets-to-market and earnings-to-price remain is not a real test.

Holding out the extreme deciles. Fit on deciles two to nine, predict one and ten. This tests extrapolation, which is what the mapping must do at the firm level: 94% of firm-months sit outside the range the portfolios span on at least one characteristic.

Subsequent returns. Do firms the mapping calls overpriced go on to underperform? This is the only criterion evaluated on firms rather than portfolios, so it is the only one that can penalise bad extrapolation.

Everything is scored on magnitudes, not only ordering: the realised wedge is regressed on the fitted one and the intercept and slope reported. A perfect mapping has intercept zero and slope one. A mapping that ranks firms correctly but compresses the spread shows a slope away from one, and would be no good for a website that reports a wedge rather than a score.

Which portfolios count. Only characteristics whose construction is verified against the replication package enter, as regressors *and* as portfolios. A portfolio we cannot reproduce carries a wedge we cannot trust, so including it adds noise to the left-hand side as well as the right. With 1 characteristic still unverified (aPM), the rule costs nothing: the leave-one-family-out fit of the direct mapping is 0.63 with every characteristic in and 0.62 without it. When six were unverified it raised that fit from 0.56 to 0.67.

Results

Leave one characteristic out

Mapping	R^2	Correlation	Intercept	Slope	RMSE (pp)
3 PCs (paper)	0.587	0.766	-0.19	0.959	6.23
10 PCs	0.747	0.864	-0.12	0.965	4.88
direct	0.766	0.875	-0.11	0.959	4.70
direct, precision-wtd	0.774	0.880	-0.35	0.944	4.63
direct + squares	0.370	0.608	-1.39	0.626	8.48
squares, precision-wtd	0.449	0.670	-1.31	0.640	8.09

Leave one family out

Mapping	R^2	Correlation	Intercept	Slope	RMSE (pp)
3 PCs (paper)	0.389	0.624	-1.68	0.665	8.15
10 PCs	0.467	0.683	-1.70	0.673	7.76
direct	0.619	0.787	-1.01	0.843	6.14
direct, precision-wtd	0.609	0.781	-1.30	0.805	6.33
direct + squares	0.038	0.195	-4.89	0.047	39.54
squares, precision-wtd	0.049	0.222	-4.88	0.056	37.44

Extremes held out

Mapping	R^2	Correlation	Intercept	Slope	RMSE (pp)
3 PCs (paper)	0.737	0.858	+1.02	0.963	7.23
10 PCs	0.805	0.897	+0.04	0.863	6.46
direct	0.804	0.897	-1.35	0.853	6.56
direct, precision-wtd	0.808	0.899	-1.37	0.856	6.50
direct + squares	0.571	0.756	+0.11	0.798	9.51
squares, precision-wtd	0.626	0.791	-1.52	0.699	9.72

Do the wedges predict returns?

Firms are sorted into deciles on their fitted wedge each month and tracked forward. Decile 1 is the most negative wedge (most underpriced), decile 10 the most positive, so if wedges are mispricing that resolves, decile 1 should outperform and the spread should be positive.

Mapping	Horizon	Weighting	Decile 1	Decile 10	Spread (pp)
3 PCs (paper)	12m	cap	18.4	10.7	7.7
3 PCs (paper)	36m	cap	60.3	34.8	25.5

Mapping	Horizon	Weighting	Decile 1	Decile 10	Spread (pp)
3 PCs (paper)	60m	cap	118.3	69.4	48.9
3 PCs (paper)	12m	equal	26.3	11.5	14.8
3 PCs (paper)	36m	equal	80.4	35.1	45.3
3 PCs (paper)	60m	equal	143.2	67.0	76.2
10 PCs	12m	cap	21.3	10.5	10.7
10 PCs	36m	cap	71.1	33.4	37.7
10 PCs	60m	cap	134.3	65.6	68.8
10 PCs	12m	equal	27.2	10.5	16.7
10 PCs	36m	equal	84.1	31.6	52.6
10 PCs	60m	equal	150.0	62.3	87.7
direct	12m	cap	18.9	10.1	8.8
direct	36m	cap	64.6	31.9	32.7
direct	60m	cap	125.1	62.4	62.8
direct	12m	equal	25.5	10.0	15.5
direct	36m	equal	80.4	30.2	50.2
direct	60m	equal	145.9	58.9	87.0

The wedges predict returns strongly and with the right sign. Over five years the direct mapping separates the extreme deciles by 63 percentage points capitalisation-weighted and 87 equal-weighted, and its deciles are monotone except capitalisation-weighted at 36 months. Three principal components produce a smaller spread (49 and 76 points) and are not monotone capitalisation-weighted at 12, 36 and 60 months.

Can the mapping be improved?

Precision weighting: no, and the reason is informative

If some portfolio wedges were estimated far more precisely than others, the regression should weight them accordingly. They are not. Reproducing the package's own standard error — a Newey–West variance on the cohort-level ratios carried through the logarithm by the delta method — and separately a moving-block bootstrap with the package's 180-month blocks:

	Median	10th pct	90th pct	Max
Standard error (pp)	29.0	27.1	32.2	40.6

The ratio of the 90th to the 10th percentile is 1.2. Precision is close to uniform across the 560 portfolios, so weighting by it has almost nothing to exploit — and in the event it is very slightly worse on all three criteria, presumably because the weights add estimation noise of their own.

What the exercise does reveal is the *level*: a standard error around 29 percentage points on wedges whose cross-sectional standard deviation is about 10. Individual portfolio wedges are imprecise, because 463 formation cohorts that overlap in 179 of their 180 months are worth only two or three independent observations. That is a caveat on the whole exercise rather than on any mapping.

Non-linear terms: no

Adding squared ranks roughly halves out-of-sample fit and pushes the slope to 0.6 or below. The design already carries the non-linearity that matters: using all ten deciles rather than the extremes lets the wedge be non-linear in each sorting characteristic, and squaring 56 standardised ranks mostly adds collinear regressors for 560 observations to overfit.

Shrinkage: yes, and it is the substance

Unpenalised, the direct regression fits best in sample and generalises worst. Penalised at the level chosen inside each fold, it is the best mapping on every criterion at once and is close to unbiased in magnitude. The penalty is not a technicality: without it the fitted wedges of unfamiliar portfolios are more than twice too dispersed. On the leave-one-family-out test the realised wedge moves 0.42 for each point of fitted wedge with no penalty, against 0.89 with the penalty of 100 (run of 12 September 2026, `group_cv.held.out` on the verified portfolios).

Number of components

Ten principal components come close to the direct regression and beat three decisively on every criterion. If a dimension-reduced mapping is wanted for interpretability, ten is the defensible choice; three is not.

Is the cross-section of wedges stable over time, and which sample should the site use?

Every portfolio wedge is $-\log$ of the mean, over formation cohorts, of the discounted fifteen-year cash flow per dollar of price. The full available sample has 559 cohorts, formed 196407 to 201101, with cash flows to 202512. Cohorts overlap in 179 of their 180 months, so the sample holds about three independent fifteen-year windows, and any subsample of it one or two. The question is whether the cross-section of wedges is the same object in different parts of the sample, and if not, which part the site should rest on.

`stability.py` re-estimates all 570 decile wedges on subsets of the cohorts: the paper's own window, halves, thirds, and every window that starts in a later year. On each subset the market price of risk is re-solved so that the market wedge is zero there, as it is on the full sample; without that, a subsample's wedges shift as a block with the market's realised return. All comparisons below use the 560 portfolios of the verified characteristics.

Each subsample against the full sample

Cohorts formed	N	λ	sd	noise	corr	rank	slope	RMSE	LS corr	flips
full 1964-2011	559	3.53	9.3	3.4	1.00	1.00	1.00	0.0	1.00	0
paper 1964-2002	462	3.97	10.0	3.4	0.99	0.99	1.06	1.4	0.99	1
first half 1964-1987	282	4.52	10.7	3.5	0.91	0.90	1.04	4.6	0.94	5
second half 1988-2011	277	2.86	10.3	3.4	0.76	0.76	0.83	6.9	0.78	8
third 1964-1979	186	5.58	12.1	3.0	0.83	0.81	1.08	7.3	0.85	8
third 1980-1995	192	2.49	5.6	3.2	0.17	0.18	0.10	10.9	-0.02	29
third 1996-2011	181	2.85	15.0	4.1	0.76	0.78	1.23	10.2	0.80	5

Cohorts formed	N	λ	sd	noise	corr	rank	slope	RMSE	LS corr	flips
from 1975	433	2.62	8.2	3.4	0.81	0.79	0.71	5.5	0.87	4
from 1980	373	2.64	8.6	3.1	0.76	0.75	0.70	6.3	0.80	7
from 1985	313	2.78	9.3	3.4	0.75	0.74	0.74	6.7	0.78	9
from 1990	253	2.90	11.3	3.2	0.77	0.77	0.93	7.4	0.79	5
from 1995	193	2.92	14.3	3.8	0.76	0.78	1.17	9.6	0.79	6
from 2000	133	2.16	13.9	4.4	0.52	0.53	0.77	12.2	0.49	11

sd: cross-sectional standard deviation of the subsample’s wedges, pp. *noise*: median idiosyncratic standard error of a wedge in that subsample, pp, from a moving-block bootstrap over its cohorts with each draw demeaned across portfolios so the component common to all portfolios drops out (block length 120 cohorts, or a third of the subsample when shorter; 300 draws). *corr*, *rank*, *slope*, *RMSE*: the subsample’s wedges against the full sample’s, slope of subsample on full. *LS corr* and *flips*: the correlation of the 56 long-short spreads with the full sample’s, and how many change sign.

Two things stand out. Adding the 2003–2011 cohorts to the paper’s window changes almost nothing: correlation 0.99, one sign flip, RMSE 1.4 pp. Dropping the early cohorts is a different matter. Every window that starts in 1980 or later correlates 0.5 to 0.8 with the full cross-section, and the 1980–1995 third is nearly flat (sd 5.6 pp against 9.3) and unrelated to the rest.

The two halves against each other

The halves are independent cohorts, so this is the cleanest test. Their wedges correlate 0.45 (rank 0.46), with a slope of second on first of 0.43. Given each half’s idiosyncratic noise, two halves of a *constant* cross-section would correlate about 0.68. The change across portfolios has a standard deviation of 11.0 pp, of which noise accounts for 5.3 pp and about 8.6 pp is left, on a cross-sectional spread of 9.3 pp. 27% of portfolios move by more than two idiosyncratic standard errors. So the cross-section moves between halves by roughly its own size. That is the honest reading, with one caveat in the other direction: a block bootstrap with blocks shorter than the 180-month overlap understates sampling noise, so some of the 8.6 pp may still be sampling variation. The gap between 0.45 and 0.68 is too large for all of it to be. Judged against the *total* Newey–West standard errors, which include the component common to all portfolios, no portfolio’s change is significant (median total noise 51 pp); that statistic answers a different question, whether a single wedge is known, and the answer to that was already no.

Where the change sits

Long-short spread of each characteristic (long leg less short leg, pp) by subsample, ordered by how much it moves between the halves. 77% of characteristics keep the sign of their spread across halves; the spreads correlate 0.54 across halves.

	full	paper	1964–87	1988–2011	1964–79	1980–95	1996–2011
D2P	-16	-23	-39	40	-43	-4	61
A2ME	-20	-27	-42	22	-47	-12	40
E2P	-26	-30	-51	12	-59	-14	18
ROC	-43	-49	-61	-14	-65	2	-27
C2D	10	13	19	-15	18	7	-25
SAT	-20	-15	-3	-34	11	0	-61
ROE	26	29	38	7	37	-15	23

	full	paper	1964–87	1988–2011	1964–79	1980–95	1996–2011
CAT	-14	-8	3	-28	17	0	-53
RNA	14	16	27	-4	28	4	-3
ATO	4	0	-13	17	-26	4	22
S2P	-42	-46	-49	-20	-47	3	-43
PROF	13	17	27	1	36	-23	12
OL	-21	-16	-6	-32	8	2	-61
AOA	-22	-24	-31	-7	-33	-5	-14
sdDVOL	-27	-26	-15	-38	-7	-13	-49
C2A	26	30	32	10	34	-13	26
NOA	-2	0	5	-17	9	-27	-9
ROIC	27	29	35	13	34	-7	25
R_12_2	23	24	13	33	15	9	48
ROA	23	27	32	12	33	-10	25
sdTURN	5	1	-2	16	-14	6	17
EPS	7	4	-4	13	-17	-14	23
R_12_7	23	24	14	31	16	10	44
TAN	17	17	16	-0	12	-6	9
R_6_2	14	15	7	22	8	3	33
TNOVR	-7	-11	-13	2	-27	5	-5
DTO	-6	-8	-12	1	-18	4	-5
AT	22	20	22	9	9	-6	13
aBEME	-52	-55	-57	-44	-63	7	-68
MAXRET	-3	-6	-1	-12	-12	-6	-21
PCM	26	27	28	17	23	-15	34
dSOUT	-38	-39	-34	-44	-36	-30	-54
IVC	-11	-13	-16	-7	-22	1	-8
PM	29	26	21	30	9	-12	56
aSAT	-12	-12	-9	-18	-4	-7	-20
RETVOL	-8	-12	-7	-15	-20	-6	-27
I2A	-18	-19	-15	-23	-22	6	-38
BETA_d	-32	-37	-35	-41	-53	-8	-67
OA	-28	-27	-30	-24	-32	-21	-28
dCEQ	-22	-25	-26	-20	-34	7	-39
SIZE	-44	-44	-41	-35	-30	11	-51
SG	-17	-19	-14	-19	-20	10	-39
R_2_1	6	6	3	8	4	1	12
dSO	-34	-34	-31	-36	-31	-25	-41
S2C	-22	-21	-17	-21	-16	16	-47
SPREAD	-14	-18	-14	-18	-28	3	-35
dGS	10	11	7	11	8	6	14
DP	-25	-33	-29	-25	-51	11	-52
SUV	-6	-6	-6	-3	-5	1	-6
IPM	29	28	25	22	13	1	34
dPIA	-22	-24	-23	-21	-27	-5	-29
IDIOV	7	4	5	3	-7	-7	2
BEME	-48	-52	-48	-47	-54	-6	-70
R_36_13	-44	-46	-43	-42	-46	-20	-56

	full	paper	1964–87	1988–2011	1964–79	1980–95	1996–2011
aSIZE	-22	-22	-19	-20	-13	6	-28
Q	-45	-49	-44	-44	-52	-2	-66

The value ratios reverse between halves, the profitability spreads shrink, and momentum grows; book-to-market, Q, size, long-term reversal and the investment-type sorts are the stable core. The 1980–1995 column is flat for almost everything.

What it does to the firm-level wedges

The direct mapping refitted on each subsample’s portfolio wedges, applied to the same firm-months, against the full-sample firm wedges (sd 28 pp).

Fitted on	corr	rank	mean diff	sd diff	diff > 10	corr 2020–25	sd diff 2020–25
full 1964–2011	1.00	1.00	+0.0	0.0	0%	1.00	0.0
paper 1964–2002	1.00	1.00	+0.5	2.4	0%	1.00	2.0
first half 1964–1987	0.96	0.96	+4.8	7.5	27%	0.96	7.0
second half 1988–2011	0.92	0.91	+1.3	11.9	40%	0.91	10.9
third 1964–1979	0.91	0.90	+12.1	12.1	63%	0.90	11.0
third 1980–1995	0.44	0.42	+30.9	25.2	85%	0.28	24.1
third 1996–2011	0.90	0.89	-9.9	20.5	67%	0.90	19.3
from 1975	0.95	0.94	+5.1	9.2	35%	0.94	8.6
from 1980	0.91	0.90	+8.1	11.9	50%	0.89	11.0
from 1985	0.91	0.91	+4.6	11.5	42%	0.91	10.6
from 1990	0.92	0.91	-0.9	12.9	43%	0.91	11.9
from 1995	0.90	0.90	-7.4	18.8	63%	0.90	17.6
from 2000	0.75	0.74	+1.7	21.2	63%	0.75	18.6

The ranking of firms is robust to the sample: correlations 0.90 to 0.96 for every window except the 1980–1995 third and the post-2000 window. The magnitudes are not: refitting on the second half moves two fifths of firms by more than 10 pp, a later window more. The halves against each other at the firm level correlate 0.79 with a standard deviation of the difference of 18 pp.

Which sample

The full available sample is the headline choice. It is the only one with about three independent fifteen-year windows; every later window rests on one or two cycles, which is why the price of risk swings from 5.6 to 2.5 to 2.8 across the thirds, and the windows ending after 2008 carry a different scale (slopes 1.2 to 1.35 on the extremes). Extending the paper’s window to 2011 cohorts and 2025 cash flows is safe. Switching to a later window is not a better estimate of the same quantity: it changes the answer in the direction of the post-1990 weakness of the value ratios and the 2000 and 2008 episodes, which is regime. What the drift does argue for is showing it: the full-sample firm wedge with the two half-sample estimates as a range, or “since 1988” as a labelled alternative specification, with the statement that rankings agree at 0.9 and magnitudes differ by about ± 12 pp.

Levels: do firms realise the wedge they are assigned?

The criteria above compare mappings on portfolios. None of them can say whether the number the site would print for a firm is the right *level*, and correlation-based comparisons hide level errors entirely. This section tests levels directly. Each candidate firm wedge is treated as a characteristic: firms are sorted into deciles on it every month at NYSE breakpoints, decile 1 holding the firms called most overpriced, and each decile is held for fifteen years as an equal-weighted and as a value-weighted portfolio. The realised wedge of that portfolio, $-\log$ of the mean discounted cash flow per dollar over the formation cohorts with the same SDF and the same λ as Table 1, is then regressed on the mean wedge the mapping assigned to the decile’s members. A mapping whose levels can be trusted shows a slope of one and an intercept of zero. The equal-weighted version is the one that matters for a website that reports one firm at a time; the value-weighted version says whether the mapping respects the calibration that sets the value-weighted market wedge to zero. `realised_levels.py` computes everything below.

A reference point first. The realised wedge of the equal-weighted universe, one dollar in every firm, is -22.5 pp, against 2.3 pp for the value-weighted universe (zero by calibration up to the admissibility filter). The average firm is a small firm and is far more underpriced, under this SDF, than the average dollar. Any firm-level wedge has to reproduce that gap.

In sample: fitted and realised on all cohorts

Firm wedge	equal-weighted deciles			value-weighted deciles			mean fitted wedge	
	slope	icept	MAD	slope	icept	MAD	EW	VW
published PWshare (PC3, EW profiles)	0.64	-18.1	16.2	0.95	-8.5	8.2	-7.5	9.0
PC3, 114 extremes, EW profiles (rebuilt)	0.67	-19.7	18.0	1.02	-9.8	9.9	-6.3	9.1
direct ridge 100: VW wedges, VW profiles	0.67	-2.9	6.9	0.79	-2.8	5.6	-33.3	1.4
direct ridge 100: EW wedges, EW profiles	1.05	+5.5	5.4	0.86	+4.5	8.0	-27.2	-8.0
direct ridge 100: VW and EW stacked	0.86	-0.5	3.5	0.76	-3.8	6.8	-27.1	0.8
direct ridge 300: VW wedges, VW profiles	0.71	-3.4	5.9	0.85	-2.3	5.1	-31.1	0.8
direct ridge 300: EW wedges, EW profiles	1.20	+9.1	6.8	1.09	+10.4	8.1	-27.1	-11.8
direct ridge 300: VW and EW stacked	0.88	-0.7	4.5	0.84	-3.1	5.8	-26.9	-0.0

slope, *icept*: realised decile wedge regressed on the decile’s mean fitted wedge. *MAD*: mean absolute gap between fitted and realised across the ten deciles, pp. *mean fitted*: the mapping’s own firm-level average, equal-weighted over firm-months and capitalisation-weighted; the latter should sit near the value-weighted universe wedge. “VW wedges, VW profiles” is the mapping used everywhere else in this report; “EW wedges, EW profiles” fits the same regression on equal-weighted decile portfolios described by equal-weighted mean ranks; “stacked” fits on both sets at once, each with its own profiles.

The published firm wedges are the paper’s PC3 mapping with equal-weighted profiles, and our rebuild of that construction reproduces them (level within 1 pp). On equal-weighted deciles they have a slope of 0.64 and an intercept of -18.1 pp: the firms the paper calls fairly priced realise wedges near -25 pp, and only the most underpriced decile realises what it is assigned. Value-weighted the slope is 0.95 and the intercept -8.5 pp. The reason is mechanical and is set out below: an equal-weighted profile places the average firm at the centre of a regression whose left-hand side is the value-weighted, big-firm wedge.

The mapping used elsewhere in this report has the level right (intercepts -2.9 and -2.8 pp) but too much dispersion (slopes 0.67 and 0.79): it assigns the bottom decile about -70 pp where -42 is realised. Its centre is coherent because value-weighted wedges are paired with value-weighted profiles, but the

typical firm sits far from that centre and the regression extrapolates. Heavier shrinkage repairs the value-weighted slope and not the equal-weighted one. Fitting on both portfolio sets with matching profiles does best: direct ridge 100: VW and EW stacked reaches a slope of 0.86 and an intercept of -0.5 pp equal-weighted, with a mean absolute gap of 3.5 pp across deciles, and 0.76 and -3.8 pp value-weighted.

The last two columns are the aggregate check. Because the market wedge is zero by calibration, the capitalisation-weighted mean of any firm-level wedge should sit near the value-weighted universe's realised wedge (2.3 pp here, the admissibility filter aside). The published series gives +9.0 pp, the value-weighted mapping +1.4 pp, and direct ridge 100: VW and EW stacked +0.8 pp; a mapping fitted on equal-weighted portfolios alone overshoots the other way. Equal-weighted, the same column should sit near the equal-weighted universe (-22.5 pp): -7.5 for the published series against -27.1 for the stacked mapping.

Decile detail, in sample

Mean fitted and realised wedge by decile of the fitted wedge (pp), for the published series and for direct ridge 100: VW and EW stacked.

published	1	2	3	4	5	6	7	8	9	10
EW fitted	30	17	9	2	-4	-10	-15	-22	-29	-42
EW realised	6	-2	-11	-28	-25	-29	-30	-30	-33	-40
VW fitted	31	17	9	2	-4	-10	-15	-21	-29	-39
VW realised	24	5	6	-7	-16	-26	-25	-25	-29	-47
direct ridge 100: VW and EW stacked	1	2	3	4	5	6	7	8	9	10
EW fitted	23	6	-3	-11	-18	-25	-31	-39	-48	-67
EW realised	24	8	-3	-11	-16	-28	-33	-38	-43	-47
VW fitted	27	7	-3	-11	-18	-24	-31	-39	-48	-63
VW realised	27	3	-10	-19	-25	-28	-29	-31	-37	-45

Out of sample

Mappings fitted on the cohorts formed before 1988 and evaluated on the cohorts formed from 1988 on (the published series, fitted on all cohorts, is evaluated on the same late cohorts). Two versions: with λ held at the full-sample value, where the late cohorts' own market wedge is +14 pp value-weighted so every intercept carries that shift; and with λ re-solved on each half, so the market wedge is zero in the fitting and in the evaluation period alike.

Firm wedge	λ	EW slope	EW icept	VW slope	VW icept
published PWshare (PC3, EW profiles)	full	0.91	-7.8	1.19	-2.0
PC3, 114 extremes, EW profiles (rebuilt)	full	0.88	-1.9	1.25	+6.8
direct ridge 100: VW wedges, VW profiles	full	0.70	+15.3	0.85	+13.1
direct ridge 100: EW wedges, EW profiles	full	1.01	+22.1	0.80	+19.3
direct ridge 100: VW and EW stacked	full	0.87	+17.0	0.88	+15.0
direct ridge 300: VW wedges, VW profiles	full	0.76	+15.4	0.91	+12.9
direct ridge 300: EW wedges, EW profiles	full	1.27	+32.0	1.18	+28.3
direct ridge 300: VW and EW stacked	full	0.91	+18.2	0.99	+15.1
published PWshare (PC3, EW profiles)	per half	0.88	-27.3	1.16	-19.1
direct ridge 100: VW wedges, VW profiles	per half	0.72	-11.9	0.84	-13.0
direct ridge 100: EW wedges, EW profiles	per half	1.03	-8.5	0.82	-6.8
direct ridge 100: VW and EW stacked	per half	0.87	-12.7	0.86	-11.9

Firm wedge	λ	EW slope	EW icept	VW slope	VW icept
direct ridge 300: VW wedges, VW profiles	per half	0.77	-12.6	0.90	-13.7
direct ridge 300: EW wedges, EW profiles	per half	1.35	-0.8	1.20	-1.8
direct ridge 300: VW and EW stacked	per half	0.94	-11.3	0.97	-12.7

Out of sample the slopes survive for the stacked mapping (0.87 equal-weighted, 0.86 value-weighted with λ per half) and the level does not: with λ held fixed the intercepts are about +15 pp, the late market’s own wedge; with λ re-solved per half they are about –12 pp, because the late cohorts’ average firm realised -32 pp against a late market of zero, below what the early cohorts taught the mapping. This is the same message as the time-split test earlier: the ranking of firms carries forward, the level of the cross-section moves with the sample, and no mapping fitted on one period can know the next period’s level.

Equal and value weighting in the method

The method is value-weighted where it defines the object and equal-weighted where it carries it to firms, and the two do not meet. Where each is used:

Step	Paper	This pipeline
Price of risk λ	value-weighted market wedge set to zero	same
Decile portfolios	value-weighted, NYSE breakpoints	same
Portfolio wedges	– log of the mean over cohorts, each cohort equal	same
Portfolio profiles (regressors)	equal-weighted mean of members’ ranks (<code>nanmean</code> ; the code asks itself “should this be value weighted?”)	capitalisation-weighted mean rank; both available since 14 September
Regression across portfolios	each portfolio one observation, 114 extremes	same, 570 deciles
Firm ranks	rank within the sortable universe	rank within firms with the characteristic
Firm-level averages, Q and investment regressions	each firm-month one observation	not run here

The calibration is the identifying normalisation of the SDF and must stay: a wedge is a statement relative to the average dollar in the market. The value-weighted decile is the standard object and its wedge is a dollar-weighted quantity. The problem is the profile. A portfolio’s equal-weighted mean rank describes its average member, which for any decile is a small firm; its wedge describes its dollars, which belong to its large members. Regressing the second on the first and then evaluating at a firm’s own rank hands every firm the big-firm level, which is what the –18 pp intercept above measures. Pairing value-weighted wedges with value-weighted profiles is coherent but centres the mapping where almost no firm is, and pairing equal-weighted wedges with equal-weighted profiles is coherent and centred on the firms but throws away the dollar-weighted information the calibration is built on. Using both, each with its own profile, is the construction that passes the level test, and the capitalisation-weighted

mean of its firm wedges is the check that it still respects the calibration.

None of this changes rankings much: every candidate above ranks firms similarly (correlations 0.9 and above). It changes the number a firm is given, by 10 to 20 pp for the typical firm, and it changes what an average of firm-level wedges means in any firm-level regression.

Recommendations

For discussion; each item names what the evidence in this report supports.

1. **Signal construction.** Adopt the documented definitions where the data departs from them and the departure changes what is measured: operating accruals net of depreciation (OA, AOA); SUV and DTO as the monthly measures Table A.1 describes rather than the last trading day's value; TNOVR with contemporaneous volume; an industry classification that is stated. Keep, and document, the conventions the panels revealed that are innocuous (BETA_d over 249 trading days, sdDVOL without zero-volume days, IDIOV with $n - 1$, the day-level spread). Treat a zero-sales ratio as missing. Drop or flag aPM: its industry means are dominated by firms with near-zero sales and its long-short alpha is indistinguishable from zero, so neither its sort nor its sign is identified.
2. **Wedge estimation.** Keep the calibration that sets the value-weighted market wedge to zero. Use the full available sample as the headline (formations to 2011, cash flows to 2025) and report the half-sample range, since the cross-section moves by roughly its own size between halves; do not move to a later window. Apply the log-bias correction to the left-hand side of the mapping or not at all; it is about 1 pp.
3. **Mapping.** Replace three principal components on the 114 extremes with a penalised direct regression on all deciles; it is better on every portfolio criterion and is the only form whose penalty can be chosen out of sample. Fit it on value-weighted and equal-weighted decile portfolios together, each with profiles of its own weighting, and choose the penalty on the leave-one-family-out fit subject to the realised-level test (slope near one, intercept near zero, equal- and value-weighted).
4. **Weighting.** Make the choice explicit at each step and never pair a wedge with a profile of a different weighting. For a site that reports one firm at a time the target is the equal-weighted realised wedge of firms with those characteristics; the capitalisation-weighted mean of the firm wedges must still equal the market wedge, and that check belongs in the pipeline.
5. **Levels, not correlations.** Evaluate every firm-level series on realised wedges of fitted deciles, and report the slope, the intercept and the decile table alongside any correlation. The published firm wedges pass the correlation test and fail the level test by 18 pp.
6. **Uncertainty and range.** A portfolio wedge carries a standard error near 30 pp and the half-sample estimates differ by about 12 pp at the firm level; present firm wedges with that range, and flag firms whose characteristics lie outside the profiles' support, where the mapping extrapolates.
7. **Downstream uses.** Firm-level regressions that use the wedge as a regressor in levels (the investment- q analysis) inherit the level error of the firm wedges and should be re-run with a level-calibrated series.

8. **Open items.** The horizon (fifteen years) and the SDF are taken as given here; sensitivity to both is untested. The move to a public characteristic library (JKP) for the extension is the right way to make the signal side reproducible.